

Program 7:

Deploy a Web Application to Kubernetes using Minikube

Project Structure:

program_7/

—>nginx-deployment.yaml

Procedure:

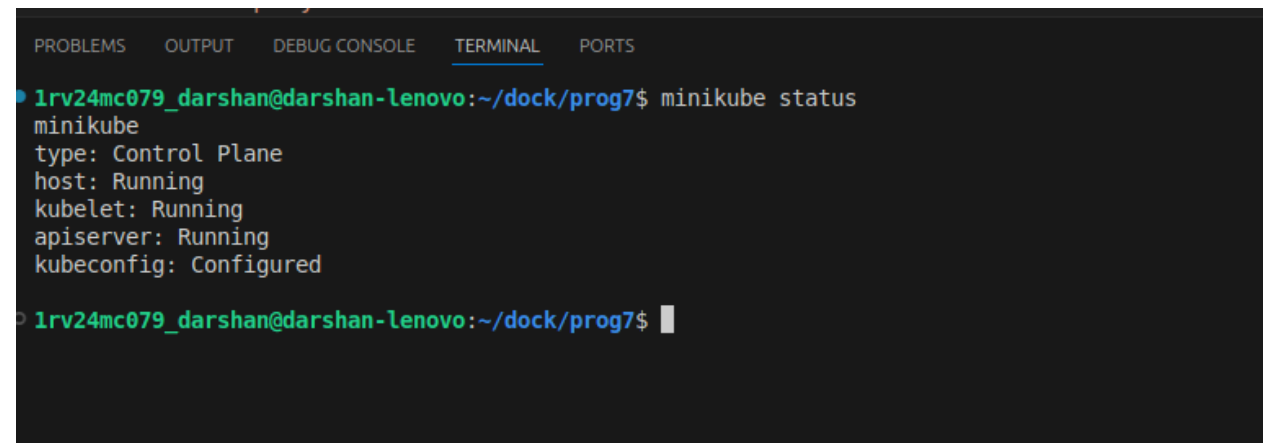
Create a folder (ex: program_7)

Step 1: Start Minikube

minikube start

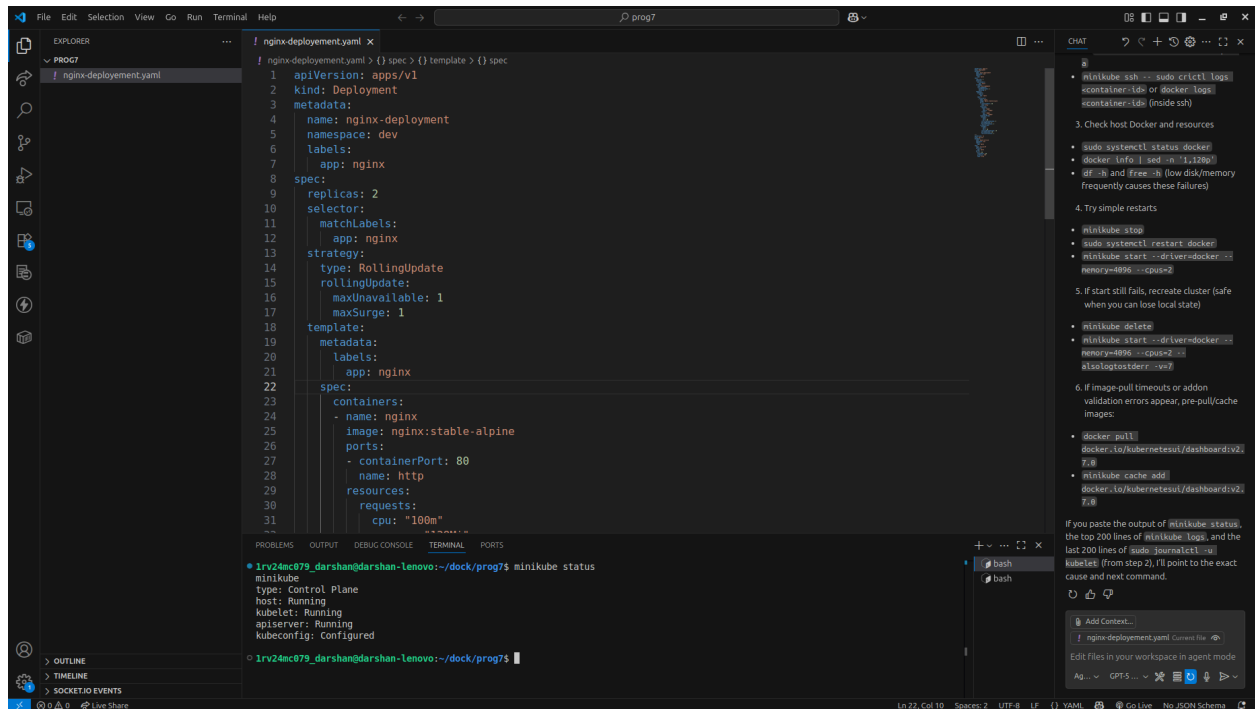
Check the status of Minikube

minikube status



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
1rv24mc079_darshan@darshan-lenovo:~/dock/prog7$ minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured
1rv24mc079_darshan@darshan-lenovo:~/dock/prog7$
```

Step 2: Create nginx-deployment.yaml file with the following code
nano nginx-deployment.yaml



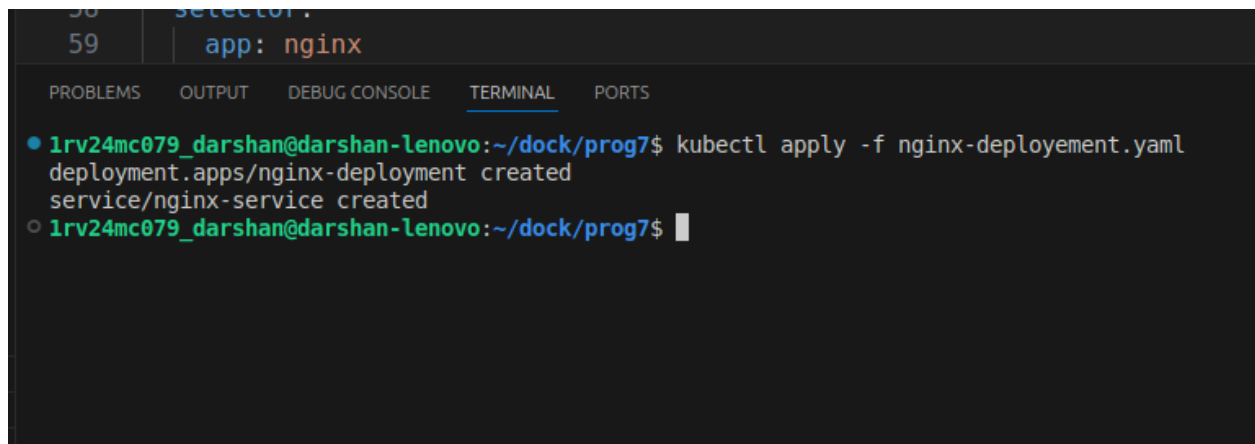
The screenshot shows the VS Code editor with the file `nginx-deployment.yaml` open. The file content is as follows:

```
1 apiVersion: apps/v1
2 kind: Deployment
3 metadata:
4   name: nginx-deployment
5   namespace: dev
6   labels:
7     app: nginx
8 spec:
9   replicas: 2
10  selector:
11    matchLabels:
12      app: nginx
13  strategy:
14    type: RollingUpdate
15    rollingUpdate:
16      maxUnavailable: 1
17      maxSurge: 1
18  template:
19    metadata:
20      labels:
21        app: nginx
22  spec:
23    containers:
24      - name: nginx
25        image: nginx:stable-alpine
26        ports:
27          - containerPort: 80
28          name: http
29        resources:
30          requests:
31            cpu: "100m"
```

The terminal window at the bottom shows the output of the `minikube status` command:

```
1rv24mc079_darshan@darshan-lenovo:~/dock/prog7$ minikube status
minikube
Type: Control Plane
Host: Running
Kubelet: Running
APIServers: Running
Kubeconfig: Configured
```

Step 3: Apply the nginx-deployment.yaml file
docker apply -f nginx-deployment.yaml



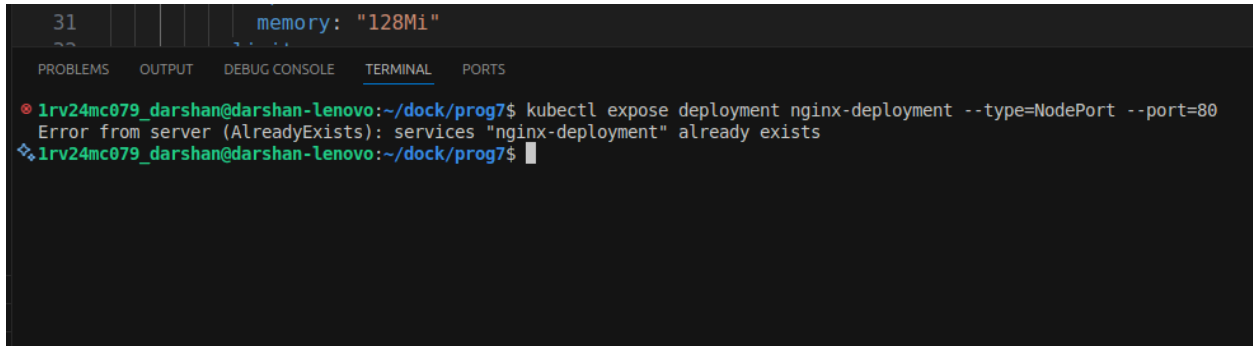
The screenshot shows the VS Code terminal with the following commands and output:

```
1rv24mc079_darshan@darshan-lenovo:~/dock/prog7$ kubectl apply -f nginx-deployment.yaml
deployment.apps/nginx-deployment created
service/nginx-service created
1rv24mc079_darshan@darshan-lenovo:~/dock/prog7$
```

Check status
docker get pods
docker get deployments

Step 4: Expose the deployment as service

kubect expose deployment nginx-deployment - -type=NodePort - -port=80



The screenshot shows a terminal window with a dark background. At the top, there are tabs for 'PROBLEMS', 'OUTPUT', 'DEBUG CONSOLE', 'TERMINAL', and 'PORTS'. The 'TERMINAL' tab is active. The terminal shows a command being executed: `kubectl expose deployment nginx-deployment --type=NodePort --port=80`. The output of the command is: `Error from server (AlreadyExists): services "nginx-deployment" already exists`. The prompt shows the user is `1rv24mc079_darshan@darshan-lenovo` in the directory `~/dock/prog7`.

Check services

kubectl get services

Acces the web app

minikube service nginx-deployment